PAUL PIOTROWSKI

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EDUCATION

B.S. Computer Science — Florida International University *May 2025 – Apr 2028*

GPA: 3.76 | Clubs: INIT, Google Developer Group, SHPE, IEEE, Code Crunch, Panther Robotics, AAVE, CASHI

TECHNICAL SKILLS

Languages & Frontend: TypeScript, JavaScript, Python, C++, Java, React 19, Next.js, Tailwind CSS, Vite

Backend, Data & AI: Node.js, FastAPI, REST APIs, Supabase, MongoDB Atlas, Gemini, Gemma, Claude, Mistral AI, Hermes

Tools & Infra: Git, GitHub Actions, GitLab, Vercel, Railway, Docker, Puppeteer/Playwright, Excel (Power Query, VLOOKUP, Pivot Tables)

WORK EXPERIENCE

Undergraduate Research Assistant — Florida International University, Miami, FL *Aug 2026 – Present*

Stack: *C# · Unity · Mixamo MCP · Local LLMs · Python · Apple Vision Pro · XR/AR*

- Develop C# and Unity modules for an XR/AR training program built for the FIU Police Department, taking features from requirement through testing and demo on the Apple Vision Pro.
- Created character animations in Unity through the Mixamo MCP pipeline, expanding the trainee scenarios the simulation supports.
- Integrated local LLM calls into the XR/AR simulation so the virtual character reacts to each trainee in real time.

Software Engineer Intern — City of Coral Gables, Coral Gables, FL *May – Jun 2026*

Stack: *Python · Gemini Veo · PyGame · OpenCV · Playwright · Raspberry Pi*

- Engineered an AI smart city kiosk proof of concept on a Raspberry Pi 5 in 3 weeks, delivering an interactive IoT installation.
- Programmed a state-driven Pygame frontend to manage the AI avatar's lifecycle, triggering Gemini Veo-generated video states (Idle, Listening, Speaking) in real time.

Stride Rent a Car — Miami, FL *Sep 2022 – May 2026*

Software Engineer *Dec 2025 – May 2026*

Stack: *React · TypeScript · Tailwind CSS · Supabase · Rently API · HQ Car Rental API · Vercel*

- Architected a 6-module enterprise dashboard integrating the Rently and HQ Car Rental APIs to manage over 1,200 vehicles and 5,000 monthly reservations, saving over 45 hours of manual work weekly.
- Launched DisputeDefender, an AI tool that extracts rental agreement data from PDFs and generates bank-ready rebuttal letters, cutting chargeback response time 90% from 30 minutes to under 3.

Sales Supervisor *Sep 2022 – Dec 2025*

- Supervised sales staff while processing customer reservations and resolving complaints for a fleet of over 1,200 vehicles.
- Handled damage claims with insurers and wrote chargeback rebuttal letters to banks.
- Compiled monthly sales and insurance reports in Excel using VLOOKUP, Power Query, pivot tables, and charts.

PROJECTS

SortArm — Winner, MLH Best Use of Tiger Data, SharkByte 2026 *Nov 2026*

Live: devpost.com/software/sortarm

Stack: *Python · LeRobot · SO-101 Arm · Feetech ST3215 · Gemini · Swarms · Tiger Data (PostgreSQL) · Tailscale · Auth0*

- Teamed with 3 engineers on a 3D-printed robot arm that sorts waste with Gemini vision and logs every sort to Tiger Data.
- Wrote the arm controller's HOME, SORT and STOP commands, with a race-proof STOP and a gripper that holds carried items.
- Secured public access through Tailscale Funnel, so anyone can watch the arm and press STOP from a public link.

UtiliTies — 2nd Place, Gridlock Challenge, ShellHacks 2026 *Sep 2026*

Live: github.com/AlexWS12/shellhacks26

Stack: *Python · FastAPI · Gemini · Jev · Next.js · PostgreSQL · MapLibre*

- Placed 2nd at ShellHacks, Florida's largest hackathon with over 1,400 attendees, by mapping over 250 real utility projects.
- Developed a multi-agent pipeline with Gemini and Jev that detects cost saving and cooperation opportunities between utilities.

ResumeFit — 1st Place, 305 SummerCodex (Build with Gemma) *Jul 2026*

Live: resume-fit-project.vercel.app

Stack: *Next.js 16 · TypeScript · Gemma (OpenRouter) · Zod · Supabase Auth · pdf-lib · unpdf · Vercel*

- Won 1st place solo in the Opportunity & Access track by building an AI tool that closes the resume loop end to end.
- Constructed a Gemma pipeline that health-checks resumes and ranks them against hundreds of real SWE internships.

CERTIFICATIONS

CodePath: AI Open Source Capstone (Aug 2026) · Technical Interview Prep, TIP102 (Apr 2026)

Anthropic: AI Fluency: Frameworks & Foundations (Jul 2026) · Claude 101 (Jun 2026)

EXTRACURRICULAR ACTIVITIES

Research Ambassador — Florida International University, Miami, FL *Sep 2026 – Present*

- Help FIU students discover and pursue undergraduate research opportunities across campus.